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Program BackwardDiff
  Implicit none
  Real::x,res
  !Interpolated value
  x=16.0
  Call BackwardDifference(x,res)
  Write(*,*)" Estimate the river's water lelvel at 4:00 pm is",res
contains
  subroutine BackwardDifference(x,res)
    Implicit none
    Real,Intent(In)::x
    Real,Intent(out)::res
    Real,Dimension(6)::time,level
    Real,Dimension(6,6)::delta
    Integer::i,j,n
    Real::u,h
    !Initialize value
    time=[2.0,6.0,10.0,14.0,18.0,22.0]
    level=[1.2,1.5,1.9,2.3,2.1,1.8]
    n=6
    delta(:,1)=level
  Do j=2,n
    Do i=n,j,-1
      delta(i,j)=delta(i,j-1)-delta(i-1,j-1)

    End Do
  End Do
  !stepsize
  h=time(2)-time(1)
  u=(x-time(n))/h
  res=level(n)
  Do j=2,n
    res=res+(u*delta(n,j))/factorial(n)
    u=u*(u+j-1)
  End Do
End subroutine
Integer Function factorial(n)
  Integer,Intent(In)::n
  Integer::i
  factorial=1
  Do i=1,n
    factorial=factorial*i
  End Do
End function

End Program

```